Chapter 10

Finite difference method in 2D

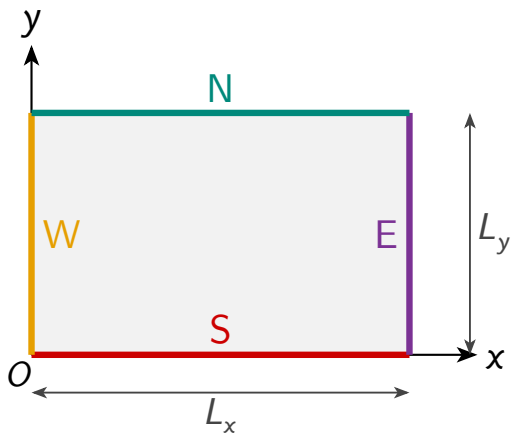
FDM 2D - problem

Hp: Cartesian coordinates, $\partial\varphi/\partial z = 0 \Rightarrow \varphi = \varphi(x, y)$; rectangular domain $L_x \times L_y$

Poisson equation: $\nabla^2\varphi = t \;\rightarrow\; \partial^2\varphi/\partial x^2 + \partial^2\varphi/\partial y^2 = t(x, y)$

Dirichlet conditions on S, W; Neumann conditions on N, E:

$$\begin{cases} \dfrac{\partial^2\varphi}{\partial x^2} + \dfrac{\partial^2\varphi}{\partial y^2} = t(x, y) & \text{inside} \\ \varphi(x, 0) = \varphi_S & \text{S edge} \\ \varphi(0, y) = \varphi_W & \text{W edge} \\ \partial\varphi/\partial x = \varphi'_E & \text{E edge} \\ \partial\varphi/\partial y = \varphi'_N & \text{N edge} \end{cases}$$



outward normal along $+x$ on E, along $+y$ on N: $\partial\varphi/\partial n = \varphi'_E,\ \varphi'_N$

FDM 2D - grid

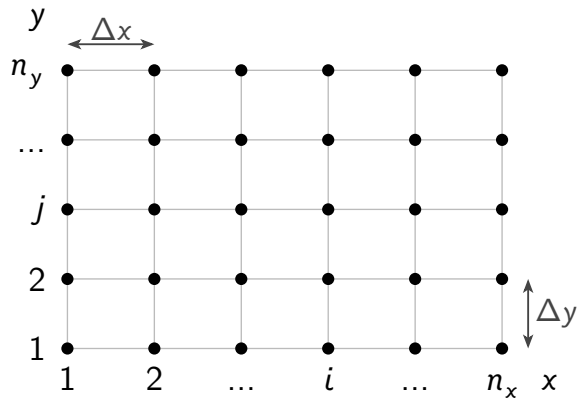
n_x nodes along x (length L_x), n_y nodes along y (length L_y)

$$\Delta x = \frac{L_x}{n_x - 1}, \qquad \Delta y = \frac{L_y}{n_y - 1}$$

Coordinates of node (i, j) (**Hp**: origin in the SW corner):

$$x_i = (i-1)\Delta x, \qquad y_j = (j-1)\Delta y$$

$\varphi_{i,j}$: approximation of $\varphi(x_i, y_j)$; $t_{i,j} = t(x_i, y_j)$



> Total number of nodes (unknowns): $n = n_x \cdot n_y$

FDM 2D - internal nodes, along x

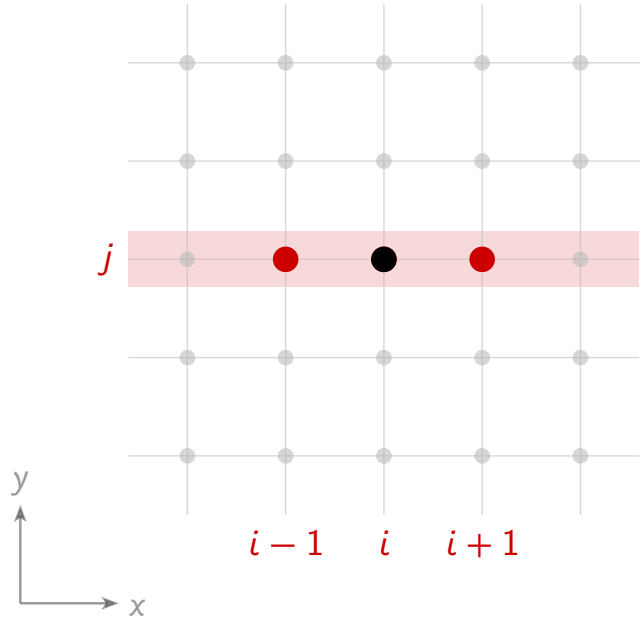
Internal nodes (i, j): $i = 2, \ldots, n_x - 1,\ j = 2, \ldots, n_y - 1$

Equation at node (i, j):

$$\left.\frac{\partial^2 \varphi}{\partial x^2}\right|_{i,j} + \left.\frac{\partial^2 \varphi}{\partial y^2}\right|_{i,j} = t_{i,j} \quad (1)$$

Along x, with j fixed (row j), as in 1D, centered formula:

$$\left.\frac{\partial^2 \varphi}{\partial x^2}\right|_{i,j} = \frac{\varphi_{i+1,j} - 2\varphi_{i,j} + \varphi_{i-1,j}}{\Delta x^2} + O(\Delta x^2) \quad (2)$$



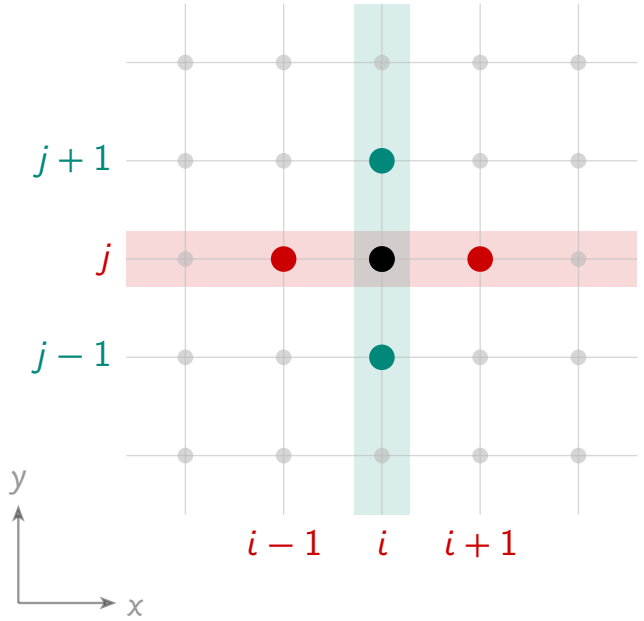
FDM 2D - internal nodes, along y

Along y, with i fixed (column i), centered formula:

$$\left.\frac{\partial^2 \varphi}{\partial y^2}\right|_{i,j} = \frac{\varphi_{i,j+1} - 2\varphi_{i,j} + \varphi_{i,j-1}}{\Delta y^2} + O(\Delta y^2) \quad (3)$$

Substituting (2), (3) into (1), neglecting $O(\Delta x^2, \Delta y^2)$:

$$\frac{\varphi_{i+1,j} - 2\varphi_{i,j} + \varphi_{i-1,j}}{\Delta x^2} + \frac{\varphi_{i,j+1} - 2\varphi_{i,j} + \varphi_{i,j-1}}{\Delta y^2} = t_{i,j}$$



row and column together: five nodes around (i, j)

FDM 2D - five-point stencil

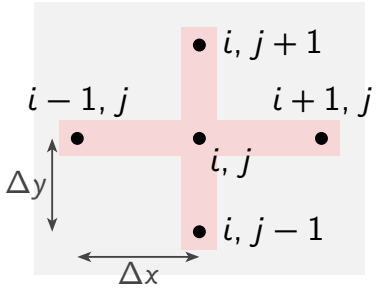
Collecting the nodal values:

$$\frac{\varphi_{i+1,j} - 2\varphi_{i,j} + \varphi_{i-1,j}}{\Delta x^2} + \frac{\varphi_{i,j+1} - 2\varphi_{i,j} + \varphi_{i,j-1}}{\Delta y^2} = t_{i,j}$$

$\downarrow$ collecting the terms in $\varphi_{i,j}$

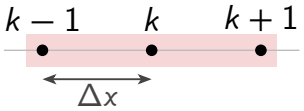
$$\frac{1}{\Delta x^2}\varphi_{i+1,j} + \frac{1}{\Delta y^2}\varphi_{i,j+1} - 2\left(\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2}\right)\varphi_{i,j}$$

$$+\frac{1}{\Delta x^2}\varphi_{i-1,j} + \frac{1}{\Delta y^2}\varphi_{i,j-1} = t_{i,j}$$



Five-point stencil: the set of nodes involved in the equation of a node

1D: three-point stencil $\varphi_{k-1}, \varphi_k, \varphi_{k+1}$



FDM 2D - harmonic case

Assume $\Delta x = \Delta y = \Delta$ and $t = 0$ (Laplace equation: φ harmonic)

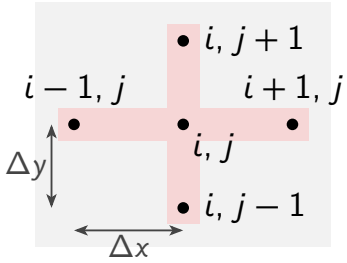
$$\frac{1}{\Delta^2}\varphi_{i+1,j} + \frac{1}{\Delta^2}\varphi_{i,j+1} - 2\left(\frac{1}{\Delta^2} + \frac{1}{\Delta^2}\right)\varphi_{i,j} + \frac{1}{\Delta^2}\varphi_{i-1,j} + \frac{1}{\Delta^2}\varphi_{i,j-1} = 0$$

$\downarrow$ multiplying by Δ^2

$$\varphi_{i+1,j} + \varphi_{i,j+1} - 4\varphi_{i,j} + \varphi_{i-1,j} + \varphi_{i,j-1} = 0$$

$\downarrow$ solving for $\varphi_{i,j}$

$$\varphi_{i,j} = \frac{1}{4}\left(\varphi_{i+1,j} + \varphi_{i,j+1} + \varphi_{i-1,j} + \varphi_{i,j-1}\right)$$



$\varphi_{i,j}$: **mean** of the values at the four neighbours
Discrete version of the mean value theorem for harmonic functions

FDM 2D - Dirichlet edges

S edge, $\varphi(x, 0) = \varphi_S$:

$$\varphi_{i,1} = \varphi_S, \quad i = 2, \ldots, n_x$$

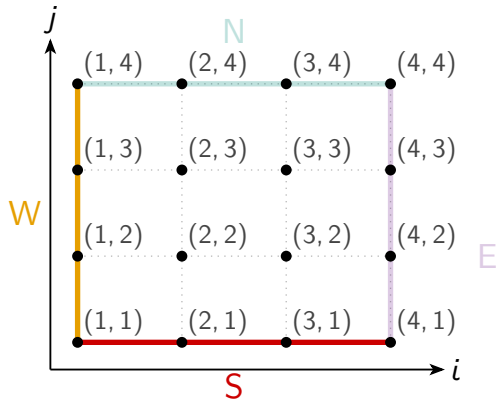
W edge, $\varphi(0, y) = \varphi_W$:

$$\varphi_{1,j} = \varphi_W, \quad j = 2, \ldots, n_y$$

both ranges start at 2 because the SW corner $(1, 1)$ is shared (below)

SW corner $(1, 1)$: on both Dirichlet edges, average

$$\varphi_{1,1} = \frac{\varphi_S + \varphi_W}{2}$$



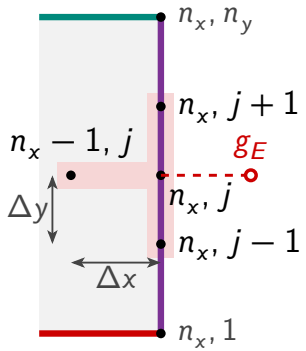
FDM 2D - Neumann condition on the E edge

Node (n_x, j), $j = 2, \ldots, n_y - 1$: the neighbour $(n_x + 1, j)$ does not exist

$j = 1$ excluded: SE corner $(n_x, 1)$, also on the S edge: Dirichlet value kept;
$j = n_y$: NE corner, later

Add a **ghost node** g_E at $x = L_x + \Delta x$, as in 1D

Two equations at node (n_x, j), both with centered formulas:



$$\begin{cases} \dfrac{\varphi_{g,E} - \varphi_{n_x-1,j}}{2\Delta x} = \varphi'_E & \text{Neumann condition} \\ \dfrac{1}{\Delta x^2}\varphi_{g,E} + \dfrac{1}{\Delta y^2}\varphi_{n_x,j+1} - 2\left(\dfrac{1}{\Delta x^2} + \dfrac{1}{\Delta y^2}\right)\varphi_{n_x,j} + \dfrac{1}{\Delta x^2}\varphi_{n_x-1,j} + \dfrac{1}{\Delta y^2}\varphi_{n_x,j-1} = t_{n_x,j} & \text{five-point stencil} \end{cases}$$

From the Neumann condition: $\varphi_{g,E} = \varphi_{n_x-1,j} + 2\Delta x \varphi'_E$

FDM 2D - E edge: ghost node elimination

Substituting $\varphi_{g,E}$ into the five-point stencil:

$$\frac{1}{\Delta x^2}\varphi_{g,E} + \frac{1}{\Delta y^2}\varphi_{n_x,j+1} - 2\left(\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2}\right)\varphi_{n_x,j} + \frac{1}{\Delta x^2}\varphi_{n_x-1,j} + \frac{1}{\Delta y^2}\varphi_{n_x,j-1} = t_{n_x,j}$$

$$\downarrow \quad \varphi_{g,E} = \varphi_{n_x-1,j} + 2\Delta x \varphi'_E$$

$$\frac{1}{\Delta x^2}\left(\varphi_{n_x-1,j} + 2\Delta x \varphi'_E\right) + \frac{1}{\Delta y^2}\varphi_{n_x,j+1} - 2\left(\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2}\right)\varphi_{n_x,j} + \frac{1}{\Delta x^2}\varphi_{n_x-1,j} + \frac{1}{\Delta y^2}\varphi_{n_x,j-1} = t_{n_x,j}$$

$$\downarrow \quad \text{collecting } \varphi_{n_x-1,j}\text{; known term } 2\varphi'_E/\Delta x \text{ to the right-hand side}$$

$$\frac{1}{\Delta y^2}\varphi_{n_x,j+1} - 2\left(\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2}\right)\varphi_{n_x,j} + \frac{2}{\Delta x^2}\varphi_{n_x-1,j} + \frac{1}{\Delta y^2}\varphi_{n_x,j-1} = t_{n_x,j} - \frac{2\varphi'_E}{\Delta x}$$

Four-point stencil for the nodes of the E edge, second-order accurate
The Neumann datum goes to the right-hand side

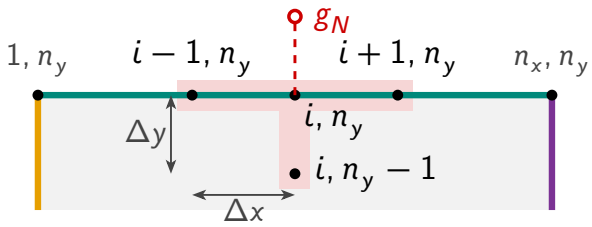
FDM 2D - Neumann condition on the N edge

Node (i, n_y), $i = 2, \ldots, n_x - 1$: same steps as for the E edge, ghost node g_N at $y = L_y + \Delta y$

$i = 1$ excluded: NW corner $(1, n_y)$, also on the W edge: Dirichlet value kept;
$i = n_x$: NE corner, next slide

Neumann condition, centered formula along y:

$$\frac{\varphi_{g,N} - \varphi_{i,n_y-1}}{2\Delta y} = \varphi'_N \quad \Rightarrow \quad \varphi_{g,N} = \varphi_{i,n_y-1} + 2\Delta y \varphi'_N$$



Substituting into the five-point stencil, collecting φ_{i,n_y-1}:

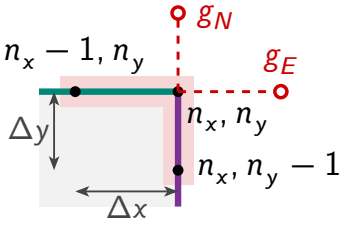
$$\frac{1}{\Delta x^2}\varphi_{i+1,n_y} - 2\left(\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2}\right)\varphi_{i,n_y} + \frac{1}{\Delta x^2}\varphi_{i-1,n_y} + \frac{2}{\Delta y^2}\varphi_{i,n_y-1} = t_{i,n_y} - \frac{2\varphi'_N}{\Delta y}$$

FDM 2D - NE corner

Node (n_x, n_y): two Neumann conditions, two ghost nodes g_E and g_N

Both conditions hold at the corner:

$$\begin{cases} \varphi_{g,E} = \varphi_{n_x-1,n_y} + 2\Delta x \varphi'_E & \text{E edge} \\ \varphi_{g,N} = \varphi_{n_x,n_y-1} + 2\Delta y \varphi'_N & \text{N edge} \end{cases}$$



Five-point stencil at (n_x, n_y):

$$\frac{1}{\Delta x^2}\varphi_{g,E} + \frac{1}{\Delta y^2}\varphi_{g,N} - 2\left(\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2}\right)\varphi_{n_x,n_y} + \frac{1}{\Delta x^2}\varphi_{n_x-1,n_y} + \frac{1}{\Delta y^2}\varphi_{n_x,n_y-1} = t_{n_x,n_y}$$

Substituting both ghost values: **three-point stencil**

$$-2\left(\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2}\right)\varphi_{n_x,n_y} + \frac{2}{\Delta x^2}\varphi_{n_x-1,n_y} + \frac{2}{\Delta y^2}\varphi_{n_x,n_y-1} = t_{n_x,n_y} - \frac{2\varphi'_E}{\Delta x} - \frac{2\varphi'_N}{\Delta y}$$

FDM 2D - linear system

One algebraic equation for each node of the grid

Assembling the equations: linear system

$$[K]\{\varphi\} = \{\text{Rhs}\}$$

$\{\text{Rhs}\}$: source t and boundary data

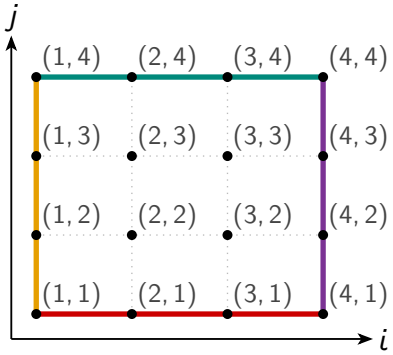
Size of $[K]$:

1D: $n_x \times n_x$
2D: $(n_x n_y) \times (n_x n_y)$ e.g. $n_x = 5$, $n_y = 4 \rightarrow 20 \times 20$

Question: the unknowns $\varphi_{i,j}$ have two indices; in which order do they go into the vector $\{\varphi\}$?

FDM 2D - linear indexing

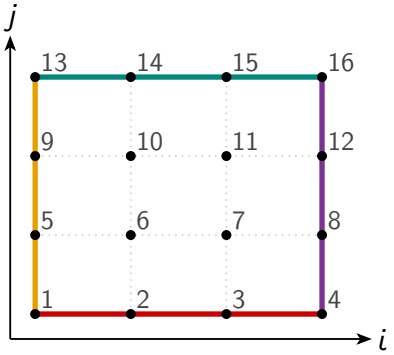
Example: $n_x = n_y = 4$, Dirichlet on S and W, Neumann on N and E



row by row:

$$\Longrightarrow$$

$$k = (j-1)n_x + i$$



Neighbours of node k: $(i \pm 1, j) \rightarrow k \pm 1, \quad (i, j \pm 1) \rightarrow k \pm n_x$ e.g. $(3,2) \rightarrow k = 4 + 3 = 7$

Coefficients of the stencil:

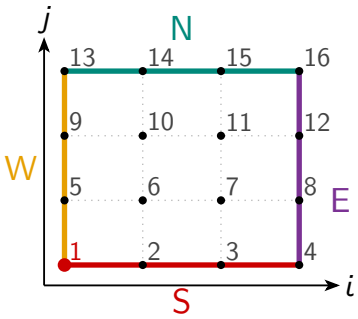
$$k_x = \frac{1}{\Delta x^2}, \quad k_y = \frac{1}{\Delta y^2}, \quad k_c = -2\left(\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2}\right)$$

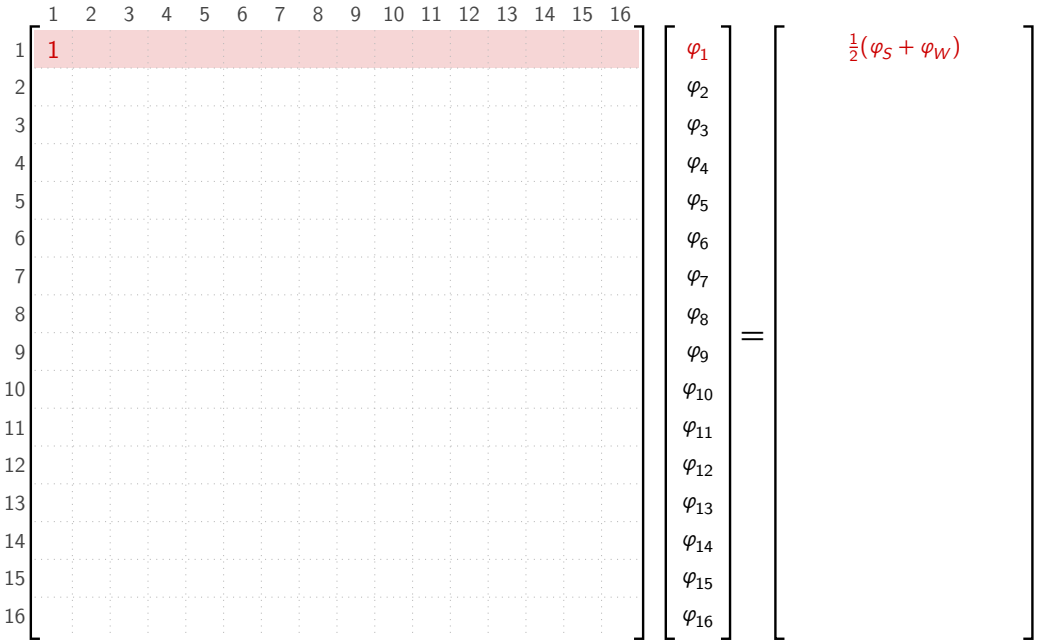
FDM 2D - assembly: node $k = 1$

$k = 1$ $(i = 1, j = 1)$: SW corner

Dirichlet on S and W: mean value

$$\varphi_1 = \frac{1}{2}(\varphi_S + \varphi_W)$$



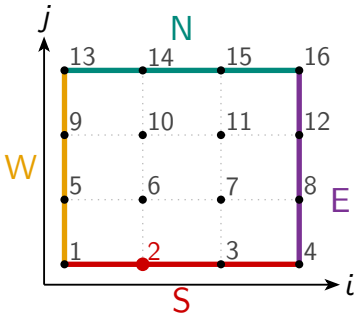


FDM 2D - assembly: node $k = 2$

$k = 2$ $(i = 2, j = 1)$: S edge

Dirichlet on S (node 3: the same)

$$\varphi_2 = \varphi_S$$



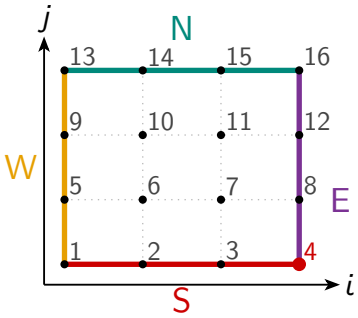
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FDM 2D - assembly: node $k = 4$

$k = 4$ $(i = 4, j = 1)$: SE corner

Dirichlet on S, Neumann on E: Dirichlet kept

$$\varphi_4 = \varphi_S$$



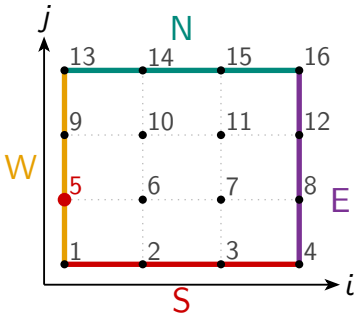
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FDM 2D - assembly: node $k = 5$

$k = 5$ $(i = 1, j = 2)$: W edge

Dirichlet on W (nodes 9, 13: the same)

$$\varphi_5 = \varphi_W$$



$$\begin{bmatrix} 1 & & & & & & & & & & & & & & & \\ & 1 & & & & & & & & & & & & & & \\ & & 1 & & & & & & & & & & & & & \\ & & & 1 & & & & & & & & & & & & \\ & & & & 1 & & & & & & & & & & & \\ & & & & & & & & & & & & & & & \\ & & & & & & & & & & & & & & & \\ & & & & & & & & & & & & & & & \\ & & & & & & & & & & & & & & & \\ & & & & & & & & & & & & & & & \\ & & & & & & & & & & & & & & & \\ & & & & & & & & & & & & & & & \\ & & & & & & & & & & & & & & & \\ & & & & & & & & & & & & & & & \\ & & & & & & & & & & & & & & & \end{bmatrix} \begin{bmatrix} \varphi_1 \\ \varphi_2 \\ \varphi_3 \\ \varphi_4 \\ \varphi_5 \\ \varphi_6 \\ \varphi_7 \\ \varphi_8 \\ \varphi_9 \\ \varphi_{10} \\ \varphi_{11} \\ \varphi_{12} \\ \varphi_{13} \\ \varphi_{14} \\ \varphi_{15} \\ \varphi_{16} \end{bmatrix} = \begin{bmatrix} \frac{1}{2}(\varphi_S + \varphi_W) \\ \varphi_S \\ \varphi_S \\ \varphi_S \\ \varphi_W \end{bmatrix}$$

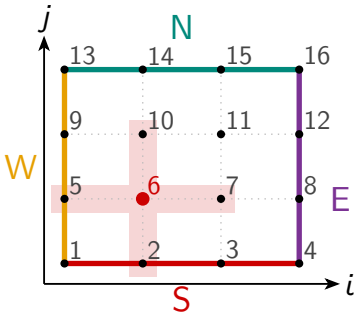
FDM 2D - assembly: node $k = 6$

$k = 6$ $(i = 2, j = 2)$: internal node

Five-point stencil:

$$k_x \varphi_7 + k_y \varphi_{10} + k_c \varphi_6 + k_x \varphi_5 + k_y \varphi_2 = t_6$$

entries in the columns $k \pm 1$ and $k \pm n_x$



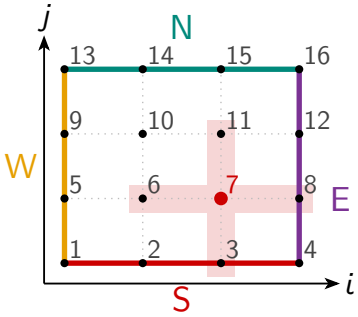
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 & & & & & & & & & & & & & & & \\
 & & & & & & & & & & & & & & & \\
 & & & & & & & & & & & & & & &
\end{bmatrix}
\begin{bmatrix} \varphi_1 \\ \varphi_2 \\ \varphi_3 \\ \varphi_4 \\ \varphi_5 \\ \varphi_6 \\ \varphi_7 \\ \varphi_8 \\ \varphi_9 \\ \varphi_{10} \\ \varphi_{11} \\ \varphi_{12} \\ \varphi_{13} \\ \varphi_{14} \\ \varphi_{15} \\ \varphi_{16} \end{bmatrix}
=
\begin{bmatrix} \frac{1}{2}(\varphi_S + \varphi_W) \\ \varphi_S \\ \varphi_S \\ \varphi_S \\ \varphi_W \\ t_6 \\ \\ \\ \\ \\ \\ \\ \\ \\ \\ \end{bmatrix}$$

FDM 2D - assembly: node $k = 7$

$k = 7$ $(i = 3, j = 2)$: internal node

Five-point stencil:

$$k_x \varphi_8 + k_y \varphi_{11} + k_c \varphi_7 + k_x \varphi_6 + k_y \varphi_3 = t_7$$



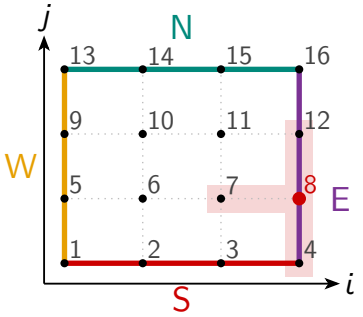
$$\begin{bmatrix}
1 & & & & & & & & & & & & & & & \\
 & 1 & & & & & & & & & & & & & & \\
 & & 1 & & & & & & & & & & & & & \\
 & & & 1 & & & & & & & & & & & & \\
 & & & & 1 & & & & & & & & & & & \\
 & k_y & & & k_x & k_c & k_x & & & k_y & & & & & & \\
 & & k_y & & & k_x & k_c & k_x & & & k_y & & & & & \\
 & & & & & & & & & & & & & & & \\
 & & & & & & & & & & & & & & & \\
 & & & & & & & & & & & & & & & \\
 & & & & & & & & & & & & & & & \\
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 & & & & & & & & & & & & & & & \\
 & & & & & & & & & & & & & & & \\
 & & & & & & & & & & & & & & &
\end{bmatrix}
\begin{bmatrix} \varphi_1 \\ \varphi_2 \\ \varphi_3 \\ \varphi_4 \\ \varphi_5 \\ \varphi_6 \\ \varphi_7 \\ \varphi_8 \\ \varphi_9 \\ \varphi_{10} \\ \varphi_{11} \\ \varphi_{12} \\ \varphi_{13} \\ \varphi_{14} \\ \varphi_{15} \\ \varphi_{16} \end{bmatrix}
=
\begin{bmatrix} \frac{1}{2}(\varphi_S + \varphi_W) \\ \varphi_S \\ \varphi_S \\ \varphi_S \\ \varphi_W \\ t_6 \\ t_7 \end{bmatrix}$$

FDM 2D - assembly: node $k = 8$

$k = 8$ ($i = 4$, $j = 2$): E edge

Four-point stencil (Neumann on E):

$$k_y \varphi_{12} + k_c \varphi_8 + 2k_x \varphi_7 + k_y \varphi_4 = t_8 - 2\varphi'_E/\Delta x$$



$$\begin{bmatrix}
1 & & & & & & & & & & & & & & & \\
 & 1 & & & & & & & & & & & & & & \\
 & & 1 & & & & & & & & & & & & & \\
 & & & 1 & & & & & & & & & & & & \\
 & & & & 1 & & & & & & & & & & & \\
 & k_y & & & k_x & k_c & k_x & & & k_y & & & & & & \\
 & & k_y & & & k_x & k_c & k_x & & & k_y & & & & & \\
 & & & k_y & & & 2k_x & k_c & & & & k_y & & & & \\
 & & & & & & & & & & & & & & & \\
 & & & & & & & & & & & & & & & \\
 & & & & & & & & & & & & & & & \\
 & & & & & & & & & & & & & & & \\
 & & & & & & & & & & & & & & & \\
 & & & & & & & & & & & & & & & \\
 & & & & & & & & & & & & & & &
\end{bmatrix}
\begin{bmatrix} \varphi_1 \\ \varphi_2 \\ \varphi_3 \\ \varphi_4 \\ \varphi_5 \\ \varphi_6 \\ \varphi_7 \\ \varphi_8 \\ \varphi_9 \\ \varphi_{10} \\ \varphi_{11} \\ \varphi_{12} \\ \varphi_{13} \\ \varphi_{14} \\ \varphi_{15} \\ \varphi_{16} \end{bmatrix}
=
\begin{bmatrix} \frac{1}{2}(\varphi_S + \varphi_W) \\ \varphi_S \\ \varphi_S \\ \varphi_S \\ \varphi_W \\ t_6 \\ t_7 \\ t_8 - 2\varphi'_E/\Delta x \\ \\ \\ \\ \\ \\ \\ \\ \end{bmatrix}$$

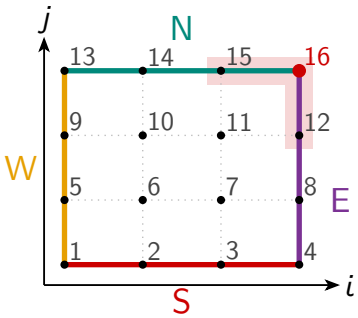
FDM 2D - assembly: node $k = 16$

$k = 16$ $(i = 4, j = 4)$: NE corner

Three-point stencil (Neumann on N and E):

$$k_c\varphi_{16} + 2k_x\varphi_{15} + 2k_y\varphi_{12} = t_{16} - 2\varphi'_E/\Delta x - 2\varphi'_N/\Delta y$$

nodes 10, 11: internal; 12: E edge; 14, 15: N edge



$$\begin{bmatrix}
1 & & & & & & & & & & & & & & & \\
 & 1 & & & & & & & & & & & & & & \\
 & & 1 & & & & & & & & & & & & & \\
 & & & 1 & & & & & & & & & & & & \\
 & & & & 1 & & & & & & & & & & & \\
 & k_y & & & k_x & k_c & k_x & & & k_y & & & & & & \\
 & & k_y & & & k_x & k_c & k_x & & & k_y & & & & & \\
 & & & k_y & & & 2k_x & k_c & & & & k_y & & & & \\
 & & & & & & & & 1 & & & & & & & \\
 & & & & & k_y & & & k_x & k_c & k_x & & & k_y & & \\
 & & & & & & k_y & & & k_x & k_c & k_x & & & k_y & \\
 & & & & & & & k_y & & & 2k_x & k_c & & & & k_y \\
 & & & & & & & & & & & & 1 & & & \\
 & & & & & & & & & 2k_y & & & k_x & k_c & k_x & \\
 & & & & & & & & & & 2k_y & & & k_x & k_c & k_x \\
 & & & & & & & & & & & 2k_y & & & 2k_x & k_c
\end{bmatrix}
\begin{bmatrix} \varphi_1 \\ \varphi_2 \\ \varphi_3 \\ \varphi_4 \\ \varphi_5 \\ \varphi_6 \\ \varphi_7 \\ \varphi_8 \\ \varphi_9 \\ \varphi_{10} \\ \varphi_{11} \\ \varphi_{12} \\ \varphi_{13} \\ \varphi_{14} \\ \varphi_{15} \\ \varphi_{16} \end{bmatrix}
=
\begin{bmatrix} \frac{1}{2}(\varphi_S + \varphi_W) \\ \varphi_S \\ \varphi_S \\ \varphi_S \\ \varphi_W \\ t_6 \\ t_7 \\ t_8 - 2\varphi'_E/\Delta x \\ \varphi_W \\ t_{10} \\ t_{11} \\ t_{12} - 2\varphi'_E/\Delta x \\ \varphi_W \\ t_{14} - 2\varphi'_N/\Delta y \\ t_{15} - 2\varphi'_N/\Delta y \\ t_{16} - 2\varphi'_E/\Delta x - 2\varphi'_N/\Delta y \end{bmatrix}$$

FDM 2D - structure of $[K]$

Non-zero entries only on **five diagonals**: 0, ± 1, $\pm n_x$

shaded: the diagonals; row-by-row numbering

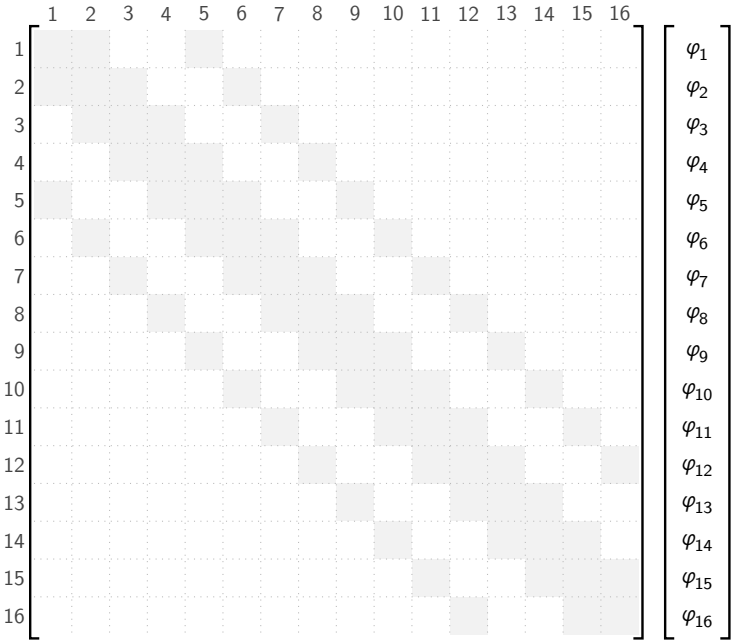
At most 5 non-zeros per row: $[K]$ **sparse**

e.g. $n_x = n_y = 100$: 10^4 unknowns, 10^8 entries, at most $5 \cdot 10^4$ non-zeros

Weakly diagonally dominant: $|k_c| = 2k_x + 2k_y$ in the stencil rows; Dirichlet rows strictly dominant

From every node, a chain of neighbours reaches a Dirichlet node

> Theorem 2 (1D chapter): $[K]$ non-singular, the discrete problem has a unique solution



FDM 2D - summary

- Grid of $n_x \times n_y$ nodes; unknowns: the nodal values $\varphi_{i,j}$
- Internal nodes: five-point stencil, $O(\Delta x^2, \Delta y^2)$; harmonic case: mean of the four neighbours
- Dirichlet: known values (average at the SW corner); Neumann: ghost nodes, four-point (edges) and three-point (corner) stencils
- Linear indexing $k = (j-1)n_x + i$; $[K]$ sparse, banded (bandwidth n_x), non-singular